

1 Product Introduction

Using an advanced one-dimensional simultaneous multi-beam system based on DBF, the radar can detect multiple types of aircraft, including multirotor UAVs, fixed-wing UAVs, FPV drones, loitering munitions, and swarms. It obtains multidimensional information in real time, including target position (range, azimuth, elevation, longitude, latitude, altitude, etc.), motion state (speed), echo intensity, RCS, and target type. The radar can accurately detect the spatial position of targets and can be cascaded with jamming, navigation spoofing, and other countermeasure devices.

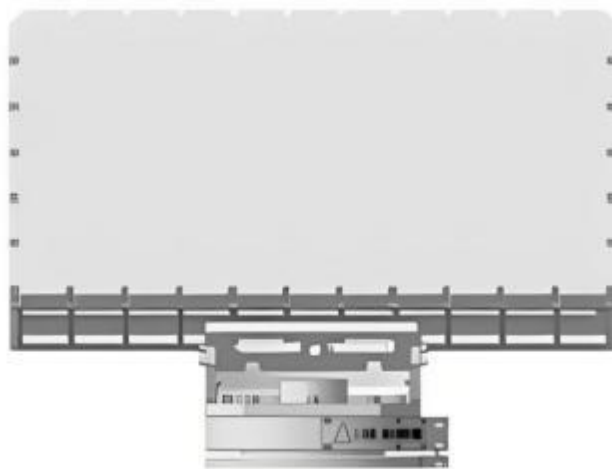


Figure 1 Low-Altitude Surveillance Radar

2 Key Features

- Adopts a solid-state active transceiver system design and digital signal processing technology, providing high reliability, safety, supportability, testability, operability, and maintainability.

- Provides omnidirectional, all-weather, round-the-clock automatic target detection and tracking capability.
- Supports normal operation at night and under rain, snow, dense fog, and other conditions.
- Uses weak-small target detection and recognition technology based on target motion characteristics, enabling effective warning, precise measurement, and data reporting for target UAVs.
- Provides detection and tracking capability for multiple batches of targets.
- Provides 360° azimuth search for blind-spot-free azimuth detection.

- Provides online monitoring, fault diagnosis, and fault location capabilities.
- Supports online loading and offline downloading of electronic maps.

Basic Specifications	
System Type	DBF
Beam Scanning Mode	Simultaneous multi-beam in elevation; mechanical scanning in azimuth
Operating Frequency Band	Ku (15.9-16.2 GHz)
Detection Capability	≥ 5 km (RCS=0.01 m ²)
Range Blind Zone	≤ 150 m
Azimuth Coverage	360°
Elevation Coverage	$\geq 60^\circ$
Velocity Measurement Range (Radial)	1 m/s - 150 m/s
Range Measurement Accuracy	≤ 7.5 m
Azimuth Measurement Accuracy	$\leq 0.3^\circ$
Elevation Measurement Accuracy	$\leq 0.5^\circ$
Data Rate	2 s
Number of Simultaneously Detected Targets	≥ 500 batches